# In vitro mass balance for PFAS: a primer from first principles

**What this is.** Everything in `MBM_Qin_vs_Armitage.md` rests on one equation and about ten approximations. This document derives the equation, names every approximation, shows where each one breaks, and then reads the five papers against it — two models and three datasets.

Read it with `scripts/mbm_published_equations.py` open. Every number quoted here comes from that file, and every code block below runs as written.

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## Part I — The one equation

### 1. What a partition coefficient actually is

Put a chemical in contact with two phases, water and something else. At equilibrium the chemical potential is equal in both:

```
mu_A = mu_B          mu_i = mu_i° + RT ln(gamma_i x_i)
```

Rearranging, the ratio of mole fractions is

```
x_A / x_B = (gamma_B / gamma_A) · exp( -(mu_A° - mu_B°) / RT )
```

The exponential is a constant at fixed temperature. So the ratio is constant **if and only if the activity coefficients *gamma* are constant**. Convert mole fractions to concentrations and you have the partition coefficient

```
K_AB = C_A / C_B                  (constant)
```

> ***Approximation 1 (dilute / linear regime).*** *gamma is constant only when the chemical is a trace. Add enough of it and the sorbent changes — sites fill, the phase swells, the chemical starts interacting with itself. Then K is no longer a constant and becomes an apparent ratio that falls as you add more. This is the single most important approximation in the whole subject. Qin's model is built around relaxing it; Armitage's is not.*

**Notation.** `K` is used for a true constant; `D` for a *distribution ratio*, which is what you measure when the chemical exists in more than one form (here: neutral and anionic) or when the sorbent is a mixture (a cell, serum). `D` is what appears in every balance below.

### 2. The well as a set of phases

A well in a 96- or 24-well plate contains, from the model's point of view:

| phase | symbol | what it is |
|---|---|---|
| medium water | V_w | the aqueous phase, where "free" concentration lives |
| serum albumin | V_alb | from the FBS in the medium |
| serum lipid | V_slip | from the FBS |
| cells | V_cell | protein + lipid + cell water |
| plastic | A_plastic | the well wall and base, a surface not a volume |

Each has a distribution ratio to water, `D_i = C_i / C_w`.

### 3. Deriving the balance

Total moles are conserved and distributed over the phases:

```
n_total = C_w·V_w + C_alb·V_alb + C_slip·V_slip + C_cell·V_cell + C_pl·A_pl
```

Substitute `C_i = D_i · C_w` in every term — this is the step that uses Approximation 1, five times:

```
n_total = C_w · ( V_w + D_alb·V_alb + D_slip·V_slip + D_cell·V_cell + D_pl·A_pl )
```

Define the **total sorptive capacity**

```
S ≡ V_w + Σ_i D_i · V_i                                  [litres of water-equivalent]
```

and the balance is a one-liner:

```
C_w = n_total / S                                        (★)
```

Everything else follows. The concentration in the cells is `C_cell = D_cell·C_w`, and the **fraction of the nominal dose that ends up inside cells** is

```
x_cell = C_cell·V_cell / n_total = D_cell·V_cell / S     (★★)
```

That is the whole model. Both papers are (★★) with different recipes for the `D_i`.

## 4. The approximation ledger

Everything that had to be true to get from the well to (★):

| # | approximation | breaks when | does it bite here? |
|---|---|---|---|
| A1 | linear sorption, *gamma* constant | sorbent saturates | **yes** — Qin measures K_d = 0.317 µM for PFOA against doses of 6.25–100 µM |
| A2 | equilibrium reached | slow uptake | no — Fragki's time course reaches steady state in ~1 h, exposure is 24 h |
| A3 | no metabolism or degradation | metabolised compounds | no for PFCA/PFSA; **yes** for fluorotelomer alcohols |
| A4 | no active transport | transporter-expressing cells | unknown — HepG2 express few PFAS transporters, but this is assumed, not shown |
| A5 | phases are independent and additive | one sorbent alters another | assumed |
| A6 | volumes additive and constant | — | fine |
| A7 | no losses (evaporation, tubing, pipettes) | volatiles | fine for ionised PFAS |
| A8 | the aqueous phase is pure water | co-solvents present | **3% ethanol / 0.5% / 0.1% DMSO** — see §6 |
| A9 | the cell is a well-mixed bag of phases | organelles, lysosomal trapping, pH gradients | assumed; Armitage's published model has no lysosome term |
| A10 | one `D_cell` describes all cells equally | different cell lines | **yes** — three assays, two cell lines, one composition used for all |

A1, A8 and A10 are the ones that have actually cost us something in this project. A4 is the one nobody can check.

## 5. Two axes, and why the choice matters

There are two ways to report the answer, and they are *not* equally robust:

```
C_cell  = D_cell · n_total / S                    [concentration]
x_cell  = D_cell · V_cell / S                     [fraction of dose]
```

`C_cell` needs the **cell volume** to convert a measured "nmol per mg protein" into a concentration. Published cell volumes for hepatic lines span 3.1× (10 ng/cell for AREc2 against 3.04 pL/cell for HepG2), and none of the three uptake papers reports its own.

`x_cell` needs only **protein per cell**, on which all sources agree (~675–694 pg/cell). So the fraction axis is the one to test models on. That is why every table in the companion memo is in % of dose.

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# Part II — Three results you can derive in two lines each

These are the levers that explain most of what we found. Each is a direct consequence of (★★); none needs a model.

## 6. Why a co-solvent almost cancels

A co-solvent makes water a better solvent, so **every** water->phase partition coefficient falls by roughly the same factor `k`. Put `D_i -> D_i/k` into (★★):

```
x'_cell = (D_cell/k)·V_cell / ( V_w + Σ (D_i/k)·V_i )
        = D_cell·V_cell / ( k·V_w + Σ D_i·V_i )
```

The `k` cancels everywhere **except against V_w**, the one term with no `D` on it. Writing `w ≡ V_w / S` for water's share of the capacity:

```
x'_cell / x_cell = 1 / ( 1 + (k-1)·w )              (★★★)
```

- `w -> 0` (long chains, albumin dominates): no effect at all.
- `w -> 1` (short chains, water dominates): the full `1/k`.

Check it against the real run. Rosenmai, Armitage, 3% ethanol ⇒ k ~ 1.35:

| | w | (★★★) predicts | model gives |
|---|---|---|---|
| PFBA | 0.80 | 1/(1+0.35·0.80) = **0.781** | 1.11/1.42 = **0.782** |
| PFNA | 0.005 | 1/(1+0.35·0.005) = **0.998** | 9.31/9.33 = **0.998** |

Exact. **An error that multiplies every sorptive term cancels on this axis. Only an error in the *relative* size of two sinks survives.**

## 7. Why albumin's share sets the sensitivity

Let `a ≡ D_alb·V_alb / S` be albumin's share of the capacity. Multiply the albumin coefficient by `f` (an error):

```
S' = S·(1 - a + f·a)        so      x'_cell / x_cell = 1 / (1 - a + f·a)   (★4)
```

For a 10×-too-low albumin coefficient (`f = 0.1`), that is `1/(1 - 0.9a)`:

| FBS | a | error in x_cell |
|---|---|---|
| 0% | 0.00 | 1.00× |
| 2% | 0.57 | 2.06× |
| 10% | 0.89 | **5.06×** |

**This is why the Fragki FBS error was so expensive.** Coding 10% where the paper says 2% does not change albumin's coefficient — it changes `V_alb` by 5×, which moves `a` from 0.57 to 0.89 and the prediction by ~4×.

It is also a warning label: *a model validated on serum-free data says nothing about its behaviour at 10% FBS.*

## 8. Why the residual's chain-length slope is a *ratio*

Once albumin dominates (a = 0.94–0.97 from PFOA up), `S ~ D_alb·V_alb`, so

```
x_cell ~ D_cell·V_cell / (D_alb·V_alb)
log x_cell ~ log(D_cell/D_alb) + log(V_cell/V_alb)
```

The second term has no chain-length dependence. Differentiate with respect to chain length n and subtract the same for the measurement:

```
d(log x_pred/x_meas)/dn  =  d log(D_cell/D_alb)_model/dn - d log(D_cell/D_alb)_true/dn   (★5)
```

**The residual's slope is the error in the slope of the ratio** — not in either coefficient alone. A model can have both coefficients wrong and no residual trend (if the errors track), or both nearly right and a strong trend (if they don't). This is exactly what separates the two models; §13 below.

# Part III — The two models

Both are (★★). They differ entirely in how they produce the `D_i`.

## 9. Armitage 2021 (IV-MBM EQP v2.0)

**Strategy: predict every `D` from one descriptor, log K_ow.**

### *9a. Ionisation*

PFAS at pH 7.4 are essentially fully anionic. Armitage splits each `D` into a neutral and an ionic part:

```
D_ow = [ F_n·10^logKow + F_i·10^(logKow - d_OW) ] / 10^(-k_salt·c_salt)
```

with `F_n` the neutral fraction, `F_i = 1 - F_n`, and `d_OW` the penalty the anion pays. The denominator is a salting-out correction, `k_salt = 0.04·logKow + 0.114` at `c_salt = 0.15 M`.

For PFAS `F_n = 0`, so every term reduces to the anionic branch. The penalties in SI Table S1:

| **phase** | **penalty `d`** |
|---|---|
| octanol | 3.1 |
| membrane lipid | 1.0 |
| **serum albumin** | **0.0** |
| plastic | 3.1 |

> ***Approximation A11.*** `d_SaW_acid = 0` *says the anion binds albumin exactly as well as the neutral form would. That is a modelling choice, not a measurement, and it is a large part of why albumin ends up holding 94–97% of the well in this model.*

### *9b. The spLFERs*

"Sorbent-phase-specific linear free energy relationships" — one regression per phase, from SI S1:

```
log K_mw = 1.01·logKow + 0.12                    membrane lipid–water
log K_pl = 0.97·logKow - 6.94                    plastic–water
log K_sa = 1.08·logKow - 0.70     (logKow < 4.5)   albumin, lower branch
log K_sa = 0.37·logKow + 2.56     (logKow >= 4.5)   albumin, upper branch
```

> ***Approximation A12.*** *log K_ow is a sufficient descriptor. For a perfluorinated surfactant with an anionic head this is the weakest link in the model: PFAS are neither simply hydrophobic nor simply lipophilic, and their K_ow is itself hard to measure.* ***Approximation A13.*** *The albumin spLFER is piecewise, flattening to a slope of 0.37 above log K_ow 4.5 — calibrated on neutral organics and pharmaceuticals, where a plateau is physically sensible because the binding cavity has finite depth. Whether PFAS–albumin binding plateaus there is an open question, and §13 shows the consequences of assuming it does.*

### *9c. The cell as a sum of phases*

```
D_cell/w = f_sl·D_ow + f_ml·D_mw + 0.035·f_prot·D_ow + f_water·1
```

Four terms: storage lipid behaves as octanol; membrane lipid as a phospholipid membrane; **3.5% of the protein behaves as octanol and the rest is inert**; the cell's own water behaves as water.

The AREc2 composition (SI Tables S3–S6), used for all three assays:

```
f_sl = 0.00046    f_ml = 0.00414    f_prot = 0.0514    f_water = 0.944
```

> ***Approximation A14, the big one.*** *"3.5% of bulk protein behaves as octanol" is a convention imported from the neutral-organics literature. Measured against Qin 2024's protein–water ratios it is* ***457–7,820× too low****. The consequence: the protein term contributes 0.1–0.2% of Armitage's cell, so the cell is 79% water at PFBA and* ***99.7% membrane lipid at PFNA****. The convention has defined protein out of the cell.*

## 10. Qin 2023

**Strategy: measure the `Ds`, and let albumin saturate.**

### *10a. The Langmuir, derived*

Take a protein with `n` identical, independent binding sites. For one site:

```
S + L <=> SL          K_a = [SL] / ([S]·[L])          K_d ≡ 1/K_a
```

Sites are conserved, `[S]_tot = [S] + [SL]`, so the occupancy is

```
theta = [SL]/[S]_tot = K_a[L] / (1 + K_a[L]) = [L] / (K_d + [L])
```

Now put it on a per-volume-of-albumin basis. The concentration of **sites inside pure albumin** is just the molar concentration of albumin times `n`:

```
C_max = n · rho_alb / M_alb = n · 1364 g/L ÷ 66,463 g/mol = n × 20,523 µM
```

So bound ligand per litre of albumin is `C_b = C_max·theta`, and the distribution ratio is

```
D_specific = C_b / C_w = C_max / (K_d + C_w)
```

Add a linear, non-saturating term for everything that is not a defined site:

```
D_BSA/w = C_max / (K_d + C_w) + D_non-specific              Qin Eq. 16   (★6)
```

**Check the derivation against Qin's own table.** He reports C_max both per litre of BSA and per mole of BSA; the ratio must be 20,523 µM:

| | C_max [µmol/L_BSA] | C_max [mol/mol] | ratio |
|---|---|---|---|
| PFBA | 9,220 | 0.450 | 20,489 |
| PFOA | 25,400 | 1.24 | 20,484 |

0.2% from the derived value. The derivation is right, and `n` = 0.45 for PFBA and 1.24 for PFOA — **less than one site for PFBA**.

### *10a-bis. How badly Approximation 1 breaks here*

A1 does not break *in* Qin's model. **Qin's model is what you build because A1 broke in the experiment** — he measured the isotherm, found it curved, and wrote (★6) to carry the curvature. Armitage keeps A1; Qin abandons it. Here is how much was at stake.

**Albumin has about one site.** From the C_max identity above, `n` = 0.45 (PFBA), 1.24 (PFOA), 0.92 (PFHxS), 0.39 (PFOS). So albumin's high-affinity capacity for PFAS is bounded at roughly **one mole per mole of albumin**.

**Now count the albumin actually in the well.** Armitage's serum albumin is 24 g/L, so the medium holds `24 × f_FBS` g/L:

| | FBS | albumin | doses | nominal PFAS : albumin |
|---|---|---|---|---|
| Rosenmai | 10% | 2.40 g/L = **36.1 µM** | 10–100 µM | 0.28 – 2.77 mol/mol |
| Amstutz | 10% | 2.40 g/L = **36.1 µM** | 100 µM | 2.77 mol/mol |
| Fragki | 2% | 0.48 g/L = **7.2 µM** | 6.25–100 µM | 0.87 – **13.85 mol/mol** |

Against `n ~ 1`, a ratio of 2.8 means roughly three times more PFAS than there are sites. A ratio of 13.9 means fourteen times. For scale, Fischer 2024 worked at <= 0.004 mol/mol and estimated the ratio in human serum at <= 0.0005 — so **these assays run 5,500–27,700× above the physiological ligand:protein ratio.** A1 is not marginally violated here; it is nowhere near true.

**What that does to the "constant".** Rearranging (★6), the specific term is just its own trace limit times a saturation penalty:

```
D_specific = C_max/(K_d + C_w) = D_trace,spec · 1 / (1 + C_w/K_d)
```

That factor `1/(1 + C_w/K_d)` **is** the activity-coefficient correction of Approximation 1, written for a site-binding sorbent. *gamma* constant means "the sorbent the Nth molecule meets is the same one the first molecule met". Fill the sites and it is not.

Between the trace limit and full saturation, `D_BSA/w` falls to the non-specific floor:

| | K_d (µM) | D at trace | D at the floor | collapse |
|---|---|---|---|---|
| PFBA | 18.3 | 546 | 43 | 12.8× |
| PFOA | 0.317 | 85,100 | 5,010 | 17.0× |
| PFHxS | 0.0744 | 256,000 | 2,950 | **86.6×** |
| PFOS | 0.00624 | 1,350,000 | 69,200 | 19.6× |

**And inside the actual wells**, solving the balance self-consistently:

| study | cmpd | dose | C_free | C_free/K_d | occupancy | D / D_trace |
|---|---|---|---|---|---|---|
| Rosenmai | PFOA | 10 µM | 0.081 µM | 0.26 | 20% | 0.81× |
| Rosenmai | PFOA | 30 µM | 0.419 µM | 1.3 | 57% | 0.46× |
| Rosenmai | PFOA | 100 µM | 5.30 µM | 16.7 | **94%** | **0.11×** |
| Fragki | PFOA | 6.25 µM | 0.367 µM | 1.2 | 54% | 0.50× |
| Fragki | PFOA | 100 µM | 22.9 µM | 72.4 | **99%** | **0.07×** |

**A parameter that is supposed to be a constant moves 7-fold inside a single dose series.** That is the whole justification for Qin's extra machinery.

**Four consequences that run through the entire comparison:**

1. **The dose exponent (§4 of the memo).** Armitage is linear by construction, pinned at b = 1.00. Measured b runs 0.85–1.42 and rises with chain length. Only a saturating model can produce that, and Qin does.
2. **Qin's error swings with dose** where Armitage's does not — his saturation is fitted to BSA isotherms, and the well's saturation need not match it.
3. **A saturating sink amplifies an error in its own size.** Cut the albumin 5× and Armitage's prediction moves by exactly the denominator arithmetic, 4.54× at every dose. Qin's moves by up to 14×, because less albumin -> more free -> higher occupancy -> lower D_BSA -> less capacity -> more free:

| Fragki PFOA, 10% -> 2% FBS | occupancy | D_BSA ratio | Qin x_cell | Armitage x_cell |
|---|---|---|---|---|
| 6.25 µM | 13% -> 54% | 0.56× | **7.89×** | 4.54× |
| 25 µM | 48% -> 93% | 0.23× | **14.17×** | 4.54× |
| 100 µM | 94% -> 99% | 0.63× | 4.56× | 4.54× |

The feedback is largest **mid-range**, where the 10% well is still linear and the 2% well has already tipped into saturation. At the top dose both are saturated and it vanishes.

4. **A literature binding constant is meaningless without its *nu*.** An apparent K_a measured at *nu* = 3 is not the number measured at *nu* = 0.004 — Qin 2024 later measured both regimes and found specific binding exceeds non-specific by 3.0–5.0×, with the changeover around *nu* ~ 0.01 to *nu* ~ 1. This is why the binding database in `ALBUMIN_BINDING.md` records the ligand:protein ratio for every row it can.

```
import sys; sys.path.insert(0, "tk_review/scripts")
import mbm_published_equations as P, mbm_qin_vs_armitage as Q

p = Q.qin_params("PFOA", 7)
trace = p["Cmax"] / p["Kd"] + 10 ** p["logDnonspec"]
a = P.ASSAYS["Fragki"]; V = P.volumes(a)
alb = P.SI["conc_ser_alb"] * a["FBS"] / Q.BSA_MW * 1e6      # uM albumin
for d in a["doses"]:
    r = P.qin_run("PFOA", d, V)
    theta = r["Cw"] / (p["Kd"] + r["Cw"])
    print(f"{d:6g} uM   nominal PFAS:alb {d/alb:5.2f}   C_free {r['Cw']:7.3f} uM"
          f"   occupancy {theta:6.1%}   D/D_trace {r['Dbsa']/trace:5.2f}x")
```

### *10b. The trace limit, and the bridge to the binding literature*

When `C_w << K_d`, (★6) collapses to a constant:

```
D_trace = C_max/K_d + D_ns = n·(rho_alb/M_alb)·K_a + D_ns
```

Dropping the non-specific term and converting to a **mass** basis (divide by *rho*_alb):

```
log D [L_w/kg_protein] = log K_a + log n - log10(M_protein/1000)          (★7)
```

For albumin (M = 66,500) the constant is **1.823**. This is the bridge that makes the entire binding literature — reported as K_a in $M^{-1}$ — comparable with a partition coefficient in L/kg. Without it, the two cannot even be compared.

> ***Approximation A15.*** *(★7) needs* `n`*, the stoichiometry, which is almost never reported with a K_a. It moves log D by up to 0.78 across the plausible range.*

### *10c. Qin's cell*

```
D_cell/w = D_SP/w · f_protein + D_lip/w · f_lipid       — and NO water term
```

`D_SP/w` is a *measured* non-specific protein–water ratio. The cell is 92–95% protein at every chain length.

> ***Approximation A16.*** *Omitting cell water means Qin's cell cannot fall below its protein term, which is why his PFBA cell partition (log 0.38) sits above Armitage's (log 0.08) — Armitage's is almost pure cell water there.*

## 11. The two models, side by side

| | Armitage 2021 | Qin 2023 |
|---|---|---|
| source of `D` | predicted from log K_ow | measured, then interpolated |
| albumin | linear spLFER, piecewise | **saturating Langmuir** |
| albumin anion penalty | 0 | n/a (measured on the anion) |

| | Armitage 2021 | Qin 2023 |
|---|---|---|
| cell | lipid + 3.5% protein-as-octanol + water | protein + lipid, no water |
| cell at PFBA | 79% water | 92% protein |
| cell at PFNA | **99.7% membrane lipid** | 95% protein |
| dose dependence | linear by construction | non-linear |
| compounds parameterised | any, from log K_ow | PFBA, PFOA, PFHxS, PFOS only |

> ***Approximation A17.*** *Qin measured four compounds. PFHxA and PFHpA are interpolated between the two carboxylate anchors (PFBA, PFOA) and* ***PFNA is extrapolated beyond both****. Only* `log D_lip/w` *has a published interpolation rule (Fig. A1,* `0.476·nCF + 0.172`*); everything else is a two-point interpolation with no basis in the paper.*

## 12. Where the dose actually goes — run this

```
import sys; sys.path.insert(0, "tk_review/scripts")
import mbm_published_equations as P, mbm_qin_vs_armitage as Q

a = P.ASSAYS["Rosenmai"]; V = P.volumes(a)
print(f"{'cmpd':7s}{'albumin':>10s}{'ser.lipid':>11s}{'water':>9s}"
      f"{'cells':>9s}{'plastic':>9s}{'  a (alb share)':>16s}")
for nm, _nC, nCF, _mw, ko, _kc, _s in Q.COMPOUNDS:
    lk = ko if ko is not None else Q.KOW_OPERA(nCF)
    D = P.armitage_D(lk, V["comp"])
    terms = {"albumin":   D["Dsa"] * V["Valb"],
             "ser.lipid": D["Dow"] * V["Vslip"],
             "water":     V["Vw"],
             "cells":     D["Dcw"] * V["Vcells"],
             "plastic":   D["Dpl"] * V["sarea"] * 1000.0}
    S = sum(terms.values())
    print(f"{nm:7s}" + "".join(f"{terms[k]/S:>{w}.1%}" for k, w in
          (("albumin",10),("ser.lipid",11),("water",9),("cells",9),("plastic",9)))
          + f"{terms['albumin']/S:>16.3f}")
```

Running it gives a = 0.196, 0.918, 0.977, 0.982, 0.971 for PFBA->PFNA, with water at 80.3% for PFBA and 0.5% for PFNA. **The balance is a water calculation for the shortest chain and an albumin calculation for everything else** — and the switch happens between PFBA and PFHxA, in one step of two $CF_2$. That abrupt switch is why PFBA behaves like a different compound in every analysis in this project, and why it has so much leverage on any regression across the series.

## 13. Why the two models fail differently

Apply (★5). Over nCF 5–8, in decades per $CF_2$:

| | albumin | cell | ratio | residual, observed |
|---|---|---|---|---|
| Armitage | +0.386 | +0.769 | **+0.383** | **+0.345** |
| Qin | +0.547 | +0.515 | **-0.032** | **-0.074** |
| Fischer 2024, *measured* BSA | **+0.540** | — | — | — |

(★5) predicts the residual slope from the ratio slope, and it does — to within the noise on a four-point fit.

**Armitage is wrong from both ends and the errors stack.** Its albumin rises at +0.386 where Fischer measures +0.540 (A13, the piecewise flattening); its cell rises at +0.769 where a flat ratio needs ~ +0.540 (A14, the cell is lipid). 40% of the gap is the spLFER, 60% the cell.

**Qin gets both right** — albumin within 0.007 of Fischer's measurement — and has essentially no chain-length residual, despite being *worse* on the level.

> ***Qin has the right chain-length physics and the wrong level. Armitage has the better level and the wrong chain-length physics.***

# Part IV — The three data papers

All three measure the same quantity on the same axis: **moles found in the cells ÷ moles added to the well**. They differ in almost everything else.

## 14. Rosenmai et al. 2018 — *J Appl Toxicol* 38:219, doi 10.1002/jat.3515

**What.** 14 PFAS in HepG2; cellular concentration by LC-MS/MS alongside a PPAR*alpha* reporter assay; the point of the paper is the *relationship* between cellular concentration and receptor activation.

**For us.** §2.5: 12-well plates, $18 \times 10^4$ cells/well, 24 h at 10/30/100 µM, 5× ice-cold PBS wash, lysates from **three wells pooled**. Table 1 is the uptake percentage — the 15 points this project uses. Recovery 78–98%.

**Gotchas.**

- Stocks are in **ethanol**, and §2.4 holds it at **3% v/v** — the largest co-solvent of the three. (§6 shows why it nearly cancels.)
- The **working volume is never stated.** 1 mL/well is an assumption.
- The 3% ethanol is stated for the PPAR*alpha* assay; §2.5 does not repeat it.
- Uptake is < 1% for everything except PFDoDA, PFTeDA and FOSA.

**Read:** §2.1, §2.2, §2.4 (the ethanol sentence), §2.5, §3.3 + Table 1.

## 15. Amstutz et al. 2022 — *Toxicology* 480:153312, doi 10.1016/j.tox.2022.153312

**What.** 13 PFAS in HepG2; MTT viability and DCFH reactive-oxygen assays at 0–800 µM, plus intracellular PFAS at 100 µM. The point is a structure–activity ranking for cytotoxicity.

**For us.** §2.5: 96-well, $4 \times 10^4$ cells, 250 µL dosed, 50 µL taken at t = 0 for a mass balance, 24 h on the remaining 200 µL, 5× DPBS. Exposure is in **the culture medium — DMEM + 10% FBS**, 0.1% DMSO carrier. Explicitly "based on the study of Rosenmai et al. (2018), with small modifications".

**Gotcha.** Amstutz reports an *intracellular fraction* normalised to **recovered** mass, not to nominal. Its denominator excludes whatever went to the plastic, so its numbers are not quite on the same axis as the other two.

**Read:** §2.2, §2.5. Skip MTT/DCFH entirely.

## 16. Fragki et al. 2022 — *Food Chem Toxicol* 172:113559, doi 10.1016/j.fct.2022.113559

**What.** A QIVIVE case study: PFOA, PFNA, PFHxS, PFOS in differentiated HepaRG, with cell-associated concentrations fed through a PBK model to oral equivalent doses. The biokinetics are a means to that end.

**For us.** The dose series in nmol/mg protein, 6.25–100 µM, Supplementary Table 2. §2.3: 24-well, 550 µL dosed, 50 µL at t = 0, 5× DPBS. The time course reaches steady state at **~1 h** — which is what licenses A2.

**Gotchas.**

- **§2.2: exposure is in "assay medium (growth medium containing 2% FBS)"**, not the 10% of the growth medium. This single sentence was worth ~4× on every Fragki prediction.
- Cells are differentiated in **1.7% DMSO for twelve days** and held there — hepatocyte-like and triglyceride-loaded, nothing like the AREc2 composition they are modelled with (A10, A14).
- The authors themselves **excluded PFNA at 100 µM** from their QIVIVE for low protein content suggesting cytotoxicity. It is the largest value in the set.

**Read:** §2.2 (last sentence), §2.3, §3.1, Supplementary Table 2.

## 17. What makes the three mutually inconsistent

| | Rosenmai | Amstutz | Fragki |
|---|---|---|---|
| cells | HepG2 | HepG2 | **differentiated HepaRG** |
| exposure FBS | 10% | 10% | **2%** |
| co-solvent | **3% ethanol** | 0.1% DMSO | 0.5% DMSO |
| plate | 12-well ×3 pooled | 96-well | 24-well |
| volume | **not stated** | 250 -> 200 µL | 550 -> 500 µL |
| normalised to | nominal | **recovered** | nominal |

And the measurements themselves disagree: at 100 µM PFHxA, Rosenmai reads 0.14% and Amstutz 0.039% — **3.6× apart on the same cell line and the same axis**. No model error explains that one.

# Part V — The same equation, in serum

Fischer et al. 2024 (*ES&T* 58:1055, doi 10.1021/acs.est.3c07415) applies (★) to serum instead of a well. Serum = water + albumin + globulin:

```
D_serum/w = VF_w + D_HSA/w·c_alb + D_glob/w·c_glob            [L_w / L_serum]
f_unbound = VF_w / D_serum/w
```

**Watch the units.** `D_HSA/w` is in L_water per **kg** of albumin, `c_alb` in kg per litre of serum, so the product is L_water per litre of serum. The protein **density cancels**: writing it as Fischer does, `D_vol·VF` with `VF = c/rho` and `D_vol = D_mass·rho`, the *rho* drops out. It survives only in `VF_w`.

This is the one place in the whole subject where the balance is *validated*: 26 PFAS have D_HSA/w, D_glob/w **and** D_serum/w measured separately, and predicting the third from the first two with no fitted parameter gives a geometric-mean bias of 1.00× and a typical error of 1.55× (`ALBUMIN_BINDING.md` §13a). The structure works. It is the in vitro *well* that does not.

# Check your understanding

**1.** In (★★), `x_cell = D_cell·V_cell/S`. If you doubled the number of cells in the well, would the measured fraction of dose in cells double? Under what condition?

**2.** The co-solvent result (★★★) says the effect scales with `w = V_w/S`. Rosenmai's PFBA has `w = 0.80`. What does that tell you about which compound you should use to *detect* a co-solvent problem experimentally?

**3.** Qin reports n = 0.45 binding sites for PFBA. A site count below one is physically odd. Give two readings of it — one that is a real property of the system, one that is an artefact of the fit.

**4.** Armitage's albumin anion penalty is `d_SaW_acid = 0` while its octanol penalty is 3.1. What would happen to albumin's share a if the albumin penalty were also 3.1, and what would that do to the model's predicted uptake?

**5.** Both models were given the AREc2 cell composition for Fragki's differentiated HepaRG. HepaRG carry visible triglyceride droplets. Which of Armitage's four cell terms would that change most, and would it make the over-prediction better or worse?

**6.** (★5) says the residual's chain-length slope is the error in the slope of `D_cell/D_alb`. Suppose someone measured a new albumin series and found a slope of +0.70 per $CF_2$ instead of Fischer's +0.540. Would that make Armitage's chain-length problem better or worse — and what would it do to Qin's?

***Answers***

**1.** Only if the cells are a negligible share of `S`. `x_cell = D_cell·V_cell/ (V_w + … + D_cell·V_cell)`. When the cell term is small the denominator barely moves and `x_cell ∝ V_cell` — doubling cells doubles uptake. When cells

hold a large share, the denominator grows too and the fraction saturates toward 1. In these assays the cell share is 0.4–18%, so the proportionality roughly holds — which is also why the *ratio* predicted/measured is insensitive to the cell count, and why §8 of the memo says the comparison survives not knowing it.

**2.** The opposite of what you would guess. The compound where water dominates is where a co-solvent shows up — so you test it with **PFBA**, not PFNA. For a long chain the effect cancels to 0.2%. Generally: to detect an error that multiplies all the `D`s, look where the one term without a `D` is large.

**3.** (a) Real: `n` is an *apparent* mean site count over a heterogeneous population of sites with different affinities; a weak, partly-occupied class of sites fitted as one homogeneous class returns `n < 1`. (b) Artefact: `C_max` and `K_d` are strongly correlated in a Langmuir fit, so if the measured concentration range does not reach saturation, the pair is only determined along a ridge and `n` is not separately identifiable. For PFBA `K_d` = 18.3 µM against doses up to 100 µM, so saturation *is* sampled — which favours reading (a).

**4.** `D_sa` would fall by 10^3.1 ~ 1,260×, so `a` would collapse from ~0.95 to near zero and the cell and plastic would carry the well. Free concentration would rise by up to the same factor and the model would over-predict uptake catastrophically. This is the sense in which `d_SaW_acid = 0` is doing enormous work: it is the assumption that makes albumin the dominant sink.

**5.** Storage lipid, `f_sl`, which is 0.00046 in AREc2 — essentially zero. Triglyceride droplets are storage lipid, so the real `f_sl` could be orders above it. That raises `D_cell`, raises `x_cell`, and makes the over-prediction **worse**. Like every other correction in this project, the direction is unhelpful — which is itself the finding.

**6.** Armitage **better**, Qin **worse**. Armitage's ratio slope is `cell - albumin` = 0.769 - 0.386 = +0.383; a steeper true albumin slope means Armitage's +0.386 is further below truth, so more of the residual is explained and a corrected albumin would flatten more of it. Qin's +0.547 would now be 0.15 *below* a true +0.70, opening a ratio-slope error it does not currently have. The point of the exercise: the ratio is what matters, so "better albumin data" helps or hurts depending entirely on which model you are holding.

# Where to go next

| file | what |
|---|---|
| `report/MBM_Qin_vs_Armitage.md` | the full comparison, §15 the medium correction, §16 the chain-length residual |
| `report/ALBUMIN_BINDING.md` §12–13a | the K_d ↔ partition bridge (★7) and the serum balance that works |
| `scripts/mbm_published_equations.py` | both models, from the published equations |
| `scripts/mbm_medium_composition.py` | §6 and §7 of this primer, run against the data |
| `scripts/mbm_chain_length_residual.py` | §8 and §13 of this primer |